Seat No. _____

003-3083001

MASTER OF SCIENCE (INFORMATION TECHNOLOGY & COMPUTER APPLICATION)(W.E.F.-2022) M.SC. (IT&CA)(2022)
Semester - 3 Examination

M.SC. (IT&CA) (2022) SEMESTER-3 OCT-2024

CS-13 FLUTTER APP DEVELOPMENT

Time : 2.5Hours] **[Total Marks : 70**

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1 (A) Answer the following: 4

1 Flutter apps use _____ programming language for creating an app.

2 _____ command checks your environment and displays a report of the status of your Flutter installation.

3 Every Dart program requires the top-level _____ function, where execution starts.

4 _____ widget arranges its children in a horizontal direction on the screen.

(B) Answer in brief (Any one out of two) 2

1 What is the use of pubspec.yaml file?

2 Explain Hot Reload feature of flutter.

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(C) Answer in detail (Any one out of two) 3

1 Explain loops in Dart with example.

2 Explain Dart libraries.

(D) Write a note on (Any one out of two) 5

1 Explain Flutter Architecture with diagram.

2 What are the Basic Flutter Widgets? Explain any two.

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2 (A) Answer the following: 4

1 _____app is one that is written in such a way that it fits the available screen size of the device it is viewed on.

2 _____widget is the animated version of Container widget that gradually changes its values over a period of time.

3 _____ method is used to navigate from one route to another route.

4 GridView is a widget in Flutter that displays the items in two-dimensional rows and columns. (True/False).

(B) Answer in brief (Any one out of two) 2

1 What is MediaQuery?

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2 How to debug UI issues in Flutter?

(C) Answer in detail (Any out of two)

1 Give difference between Stateless and Stateful widgets.

2 Explain SliverAppBar widget. 3

(D) Write a note on (Any one out of two)

1 Explain the layout and widgets hierarchy with example.

2 Explain custom widget creation in Flutter. 5

3 (A) Answer the following:

1 BloC stands for _____. 4

2 Flutter is not declarative (True/False).

3 _____ type of state can be used globally.

4 BaaS stands for _____

(B) Answer in brief (Any one out of two)

1 What are DevTools? 2

2 What is InheritedWidget?

(C) Answer in detail (Any one out of two)

1 Explain Stream-based state management with RxDart. 3

2 Write a note on ScopedModel.

(D) Write a note on (Any one out of two)

1 Explain BLoC pattern for state management. 5

2 Explain Redux architecture for Flutter.

4 (A) Answer the following:

1 Flutter provides _____ package to use http resources. 4

2 JSON stands for _____.

3 Flutter apps can make use of the SQLite databases via _____ plugin available on pub.dev.

4 REST stands for _____

(B) Answer in brief (Any one out of two)

1 What is JSON serialization? 2

2 What is Hive in Flutter?

(C) Answer in detail (Any one out of two)

1 Explain caching data in Flutter. 3

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2 Explain the use of WebSockets for real-time communication in Flutter.

(D) Write a note on (Any one out of two) 5

1 Explain Shared preferences and secure storage in Flutter.

2 How to perform CRUD operations on a SQLite database in Flutter? Explain with example.

5 (A) Answer the following: 4

1 What is tween animation in flutter?

2 _____ plugin in Flutter is used to access the device's location services.

3 FCM stands for _____

4 _____ package in Flutter helps you create Mock Dependencies of classes while testing

(B) Answer In brief (Any one out of two) 2

1 Explain Accessibility in Flutter apps.

2 Give steps to integrate camera in Flutter app.

(C) Answer in detail (Any one out of two) 3

1 Explain Push notifications in Flutter using Firebase.

2 Explain Testing and debugging in Flutter.

(D) Write a note on (Any one out of two) 5

1 Explain animation techniques – Flare and Lottie.

2 Explain Internationalisation and localization in Flutter.

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